

Lab 02 – Reservoir Stage Analysis: One-Page Results Summary

Sum-of-two-logistics model, fitted by Levenberg–Marquardt (scipy.optimize.curve_fit, method='lm') · n = 288 fifteen-minute readings, t = 0–71.75 h

1. Model & Parameter Estimates

$$h(t) = h_0 + A_1 / (1 + \exp(-k_1 \cdot (t - t_1))) - A_2 / (1 + \exp(-k_2 \cdot (t - t_2)))$$

Parameter	Estimate	Std. Error (SE)	t-statistic	p-value
h0 (baseline stage)	14.35 m	0.01 m	1483.05	< 0.001
A1 (rise amplitude)	8.92 m	0.75 m	11.84	< 0.001
k1 (rise steepness)	0.512 1/h	0.021 1/h	24.05	< 0.001
t1 (rise inflection time)	31.52 h	0.09 h	348.06	< 0.001
A2 (fall amplitude)	3.72 m	0.76 m	4.90	< 0.001
k2 (fall steepness)	0.283 1/h	0.014 1/h	20.59	< 0.001
t2 (fall inflection time)	37.55 h	1.06 h	35.51	< 0.001

All seven parameters are statistically significant (p < 0.001, two-tailed, df = n–p = 281).

2. Goodness-of-Fit Metrics

SSE	SST	R ²	s (std. error of estimate)	dof (n – p)
1.802 m ²	2034.08 m ²	0.9991	0.080 m	281 (288 – 7)

3. Calculated Area Under the Fitted Curve

Method	Value	Note
scipy.integrate.quad	1260.97 m·h	abs. error estimate = 5.09×10 ^{–7} m·h
np.trapezoid (raw data)	1261.00 m·h	cross-check on logged readings
Difference	–0.03 m·h	–0.002% of the trapezoidal value

Units are **meter-hours (m·h)** — a time-integral of stage, **not a volume** (no storage-area term is included). This value is a compact measure of how high and how long the reservoir stayed elevated during the event; it should not be used directly for spillway sizing or inflow/outflow volume calculations.

4. Residual Analysis

Mean residual ≈ 0.0000 m (globally centered), but the residuals are **not randomly distributed** — they trace a smooth, repeating wave rather than white noise scattered around zero. The longest unbroken run of same-sign residuals is **54 consecutive points (13.5 h)**, far exceeding the 1–3 points expected from a correctly specified model. Residual spread does **not** increase with water level (std. dev. = 0.088 m low third, 0.080 m mid third, 0.060 m high third of stage); error is instead concentrated in the curve-bending (transition) regions. The largest residual is **0.221 m at t = 30.25 h**, about **22×** the logger's 0.01 m resolution, confirming this is a real structural mismatch and not sensor noise. **Conclusion:** despite R² = 0.9991, the systematic residual pattern indicates a fit that fails to capture the local event shape, and R² alone should not be used as sole evidence of goodness of fit for this record.